S1 Supplementary Methods

Pinned measurement contract and analysis design

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S1.1 Dataset and contract

The analysed export is `neuralmanifolddynamics_ds004511_20260921_050009`: 134 H5 recordings from 45 people (Rest 45, CC 45, GG 44). The canonical MNPS `[m,d,e]` and stratified 9D definitions are fixed. Coordinates are `subject_anchored`; values are not re-centred or re-tuned against task contrasts. Task labels come from the BIDS-style filename stem, not within-subject order.

S1.2 Estimator separation

Occupancy is a chart-location summary. Chart-Jacobian alpha is the spectral abscissa of the local chart Jacobian. L2/L3 quantities derive from an identified lag-1 one-step map, only where support permits; L3 alpha is the spectral abscissa of the matrix-log generator proxy `log(Phi)/dt`, not of `Phi` itself. L4 amplitude summaries are finite-horizon certificate-derived summaries. These objects are reported in separate panels and never substituted for each other. Lag-2 was attempted but yielded only 13 direct L2 maps and 10 L3 proxies, too sparse and task-imbalanced for inference. `reach_d_eff` is excluded from dimensional interpretation after the degeneracy audit.

S1.3 Aggregation and inference

Each finite recording-level included median is reduced to a subject-by-task median. Pairwise contrasts are complete-case later-minus-earlier Wilcoxon signed-rank tests with paired d_z; omnibus results use complete-triplet Friedman tests. BH correction is applied across finite Wilcoxon tests separately for (i) occupancy/MNJ/L1/L4 and (ii) chart-Jacobian alpha/L2/L3. Bootstrap uses 10,000 replicates and seed 20260921. No missing value is represented as zero.

S1.4 Status and prohibited claims

This is `exploratory_discovery_20260921_g4_not_locked`, internally checked on one export. It is neither a mechanistic experiment nor a confirmatory test. Subject anchoring precludes interpreting amplitudes as between-subject levels; partial lag-1 identification defines the eligible sample rather than a recoverable nuisance.

Bibliography